

Addressing Statistical Heterogeneity in Federated Learning for Seaship Datasets*

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Abstract—The Federated Machine Learning (FedML) paradigm was introduced to protect raw data sharing while training a collaborative model that incorporates the knowledge from the data generated at the source. Training FedML models is straightforward for homogeneous clients (that shares the class and feature spaces) across clients. However, real-time datasets are heterogeneous in terms of label and feature distribution. Heterogeneity stems from (a) skewed label and (b) skewed feature distribution. In the former, the number of classes across the clients varies, while in the latter, the same label can be associated with different features across the clients (eg, clients capturing boat images from different angles). There is also a third class of problem, which is similar features but different labels. This class of problem stems from discrepancies in the granularity of labeling strategies employed for annotated datasets. That is, a client may identify all classes of boats (sailboat, fishing vessel, canoe) as one class called “Boats”, while another client identifies the boats at fine-grained granularity.

In this project, we address heterogeneity in both inter-client (similar features but different class labels within a single client) and intra-client (dissimilar features for the same label across clients). To overcome the challenge of inter-client heterogeneity, we propose the implementation of clustering techniques and feature extraction on datasets, including ABOShips [1], Seaships [2], VIS Offshore, and VIS Onshore Image Dataset. Here, we extract the features of each class and cluster the classes that establish similarity in the feature space. We address the intra-client heterogeneity similarly by clustering the clients based on their similarity in feature space for the same label. We demonstrate the performance of our clustering strategies on the FedML platform, which allows us to simulate a server and clients that represent multiple edge devices.

Index Terms—Federated Learning, Statistical heterogeneity, Maritime Environments

I. INTRODUCTION

In today’s era where data is easily accessible, data privacy is a primary concern in preventing the leakage of private information. Traditional machine learning requires data sharing across borders to train models with high accuracy. The more data that is used to train the model, the more accurate the model becomes. However, creating an accurate model without sacrificing users personal data and privacy is no small challenge.

Thus the federated learning paradigm was created to protect user’s data, yet train a collaborative model that is able to

incorporate the knowledge from the data generated at the source. Federated learning models are able to protect their users data by not strictly extracting the information from the users. Opposed to traditional machine learning models that take all the user data directly, the federated learning model takes it more indirectly.

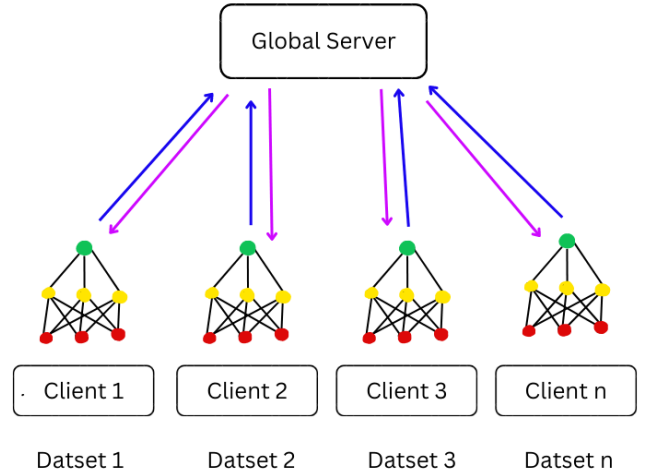


Fig. 1. Federated Learning Model Architecture

Traditional machine learning models compile long lists of user data and are able to feed this data directly into the model to train it and update the model’s weights. Instead of taking the data itself, the federated learning paradigm pushes individual client models to the users and allows them to update based on the user’s data. These client models are called local models which push their updated weights to the global model in batches. The global model is a model that is able to aggregate all of the weights together by performing federated averaging. Then, the global server is able to perform accuracy tests. The disadvantage to federated learning is that on average compared to traditional machine learning, the federated learning model requires more data to train to the same accuracy. Since the model is being updated in smaller batches and it is the weights themselves, the process takes more batches and averaging to converge.

Homogenous clients are clients that share the same names

This work has been partially supported by ONR Grant #N00014-23-1-2119, U.S. Army Grant #W911NF2120076, NSF CAREER Award #1750936, NSF REU Site Grant #2050999 and NSF CNS EAGER Grant #2233879.

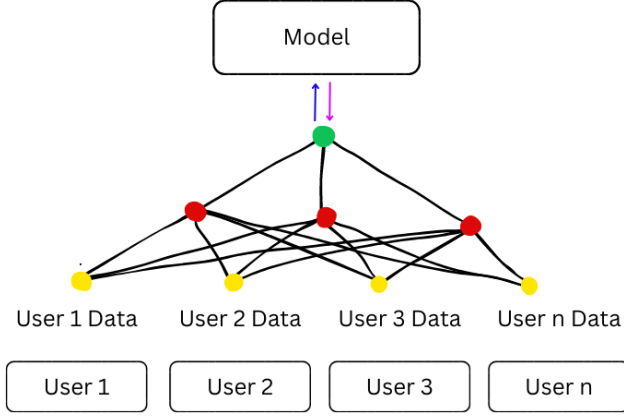


Fig. 2. Traditional Machine Learning Model Architecture

and feature classes. Training a federated learning model is straightforward for homogeneous clients because the data annotations are all done to the same granularity. However, real time datasets are heterogeneous in terms of labels and feature distribution. The images will differ based on the decided granularity and labels/features the individual annotators decided upon. In order for the model to be able to accurately and efficiently classify and detect ships from images, it is necessary to train it with multiple datasets and images of ships from all over the world. For this reason we must find a way to combine many heterogeneous datasets into homogenous datasets.

Addressing the issue of heterogeneity in a federated learning setting is a formidable challenge. Heterogeneity stems from (a) skewed label distributions and (b) skewed feature distributions. The number of classes across clients vary and the same label can be associated with different features. It is also possible for the datasets to have similar features but different labels. Depending on the granularity of labeling, one dataset may label a class much more specifically than another. When training a model with multiple heterogeneous datasets, to prevent model corruption we must homogenize the datasets with a set granularity and assure the proper labels correspond to their matching features within the scope of the problem.

Although we applied this to seaships datasets for naval implementation, it is not restricted to use in ship detection and classification. This framework can be applied to preparing datasets with common features to be used to train a model. It can be used across multiple edge devices in a federated learning setting to homogenize the users data and efficiently and accurately train a model.

II. RELATED WORK

Current federated learning challenges include communication efficiency, data heterogeneity, computational constraints, privacy and security, and system complexity. There are two

commonly used application settings for the federated learning paradigm: cross-device and cross silo. Specific to data heterogeneity the paper explains how local datasets on devices can have many different distributions and sizes. Consequently local objectives utilizing the empirical risk minimization (ERM) objective function can be different if different user models have arbitrarily different local minima. When performing local updates from the same global model, clients end up drifting towards the minima of local objectives end end up with different local models. Further, computational heterogeneity including differences in compute speeds can cause and increase in the runtime of an algorithm allowing stragglers to be discarded. To address the issues of data heterogeneity the paper recommends modifying the client weighting strategy to account for variability. [3]

Related work addresses the statistical heterogeneity by focusing on the model weights to optimize convergence. They address model aggregation in scenarios where local training data distribution varies across clients. The authors mention that with Federated Averaging, the model suffers degradation when faced with data that is non identically distributed causing the local models to diverge from the global objective. This approach utilizes the spectral decomposition of the Hessian of local loss functions to assign unique weight coefficients to each parameter update in the aggregation process. With the method they propose, the model converges faster than federated averaging and addresses the “client drift” piece of statistical heterogeneity. [4]

A concern discussed by another paper is the issue of resource waste, straggler issues, and model bias from heterogeneity within the hardware resources and local datasets. Their approach involves tracking and adapting to client patterns to reduce bias during the training process due to resource heterogeneity. The authors propose a scheduling algorithm so that the model can measure performance and bias and calculate the client availability as well as resource usage. It is able to combine both local and global accuracy measurements. [5]

Another approach clusters the clients themselves on specialized local aggregation servers to address the heterogeneous data across clients. The approach is called AdaCFL which classifies clients into clusters based on their local data distribution and it is able to train a specialized model for each cluster. It also explains how federated learning is a good solution for analyzing data stored on edge nodes while maintaining privacy. Different from Federated Averaging and Clustered Federated Learning, AdaCFL clusters the clients themselves. [6]

A survey on ship detection explores the concept of utilizing a Feature Pyramid Network (FPN) to extract the features of the images. The use of feature extraction is to address the challenge of detecting objects that are different sizes within an image. When the object varies in scale within an image it can be difficult to use a single fixed-size receptive field to capture the information. By using an image pyramid it is possible to achieve multiscale feature detection; however, this approach requires significant computational power. The FPN is able to efficiently generate multiscale features within

a neural network architecture. It is able to combine high level semantic information with lower-level feature maps that have finer details so the network can capture objects of different sizes and improve object detection. One disadvantage to it is its capabilities of detecting small objects. Another extended version (EFPN) was proposed to address the issue. [7]

III. OVERVIEW/OVERALL FRAMEWORK

In this paper we propose a framework to homogenize heterogeneous datasets for training a federated model and determine necessary granularity for accurate model performance. We utilize the open source ship datasets including: ABOShips [1], SeaShips [2], as well as VIS Offshore and Onshore for model implementation in a naval setting. We aim to address heterogeneity in both inner-client and intra-client. Inner-client heterogeneity is where the clients have similar features but different class labels within a single item. Intra-client heterogeneity is where there are dissimilar features across clients but they are given the same label. We address both forms of heterogeneity by implementing clustering techniques as well as feature extraction within the datasets. By extracting the features of each class and clustering based on the feature similarity we are able to determine how similar the images are to each other without using the annotation and helping eliminate inaccuracy and heterogeneity in the labels. We are able to demonstrate the performance of our clustering strategies on the FedML [8] platform which allows us to simulate a server and clients that represent multiple edge devices.

IV. METHODOLOGY/APPROACH

To prepare the dataset to train a model for object detection, we begin by configuring a python environment. For hardware, we utilized a Ubuntu 18.04 Linux PC as well as an Ubuntu 20.04 laptop. We configured new users and installed Anaconda as well as FedML and necessary dependencies for the two softwares. We utilized the python environment to perform cropping, feature extraction, and create the t-sne plot. Some software dependencies for the python code include: img2vec-pytorch PIL (python imaging library) pytorch tensorflow numpy sklearn matplotlib plotly resnet torchvision scikit-learn Pillow open cv ast (abstract syntax trees) standard python library (os, csv).

We configured the FedML MLops platform and were able to individually bind each of the devices as clients. We were able to test the FedML platform by running the FedML Octopus Example with MNIST + Logistic Regression simulation with the laptop and pc as two clients on one server.

We downloaded the following open source datasets: SeaShips, ABOShips, and VIS onshore and offshore. SeaShips has 6 classes of ships: bulk cargo carrier, container ship, fishing boat, general cargo ship, ore carrier, passenger ship. ABOShips has 11 classes of ships: seamark (37.89%), boat (20.58%), sailboat (38.88%), motorboat (41.11%), passengership (26.71%), cargoship (1.58%), ferry (9.56%), miscboat (28.30%), miscellaneous (1.30%), militaryship (25.90%), cruiseship (13.63%).

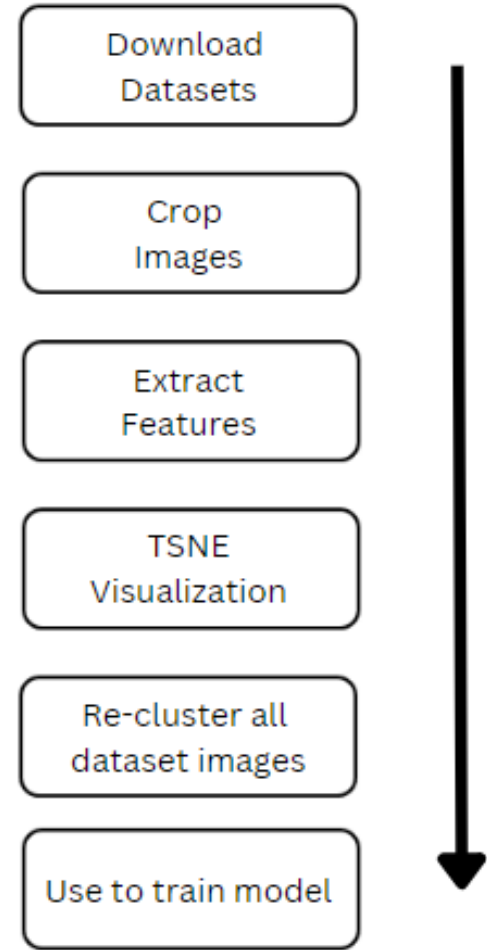


Fig. 3. Process of Dataset Preparation

Since the images were all taken from different parts of the world and are separated into different categories based on different features where only some overlap, we must determine the true categories that the ships can be combined into at a fixed granularity for training a model.

To do this, we begin by cropping each image. Each image within the dataset is annotated by pixel for the x-min, x-max, y-min, and y-max for where there are ships within the images. They are also annotated based on the ship classes. The information is given in a separate file therefore we wrote code in python to crop the images and sort them into their folders based on their ship class. We did this for both SeaShips as well as ABOShips.

We utilized an off the shelf feature extractor called img2vec. We cloned the git repository and modified the code as needed. To test the feature extractor we ran the simulation that classified humans and cats. Since the code required an input for the number of classes we were not able to use the code to classify the ships since part of the research question is determining the number of ship classes to separate the images

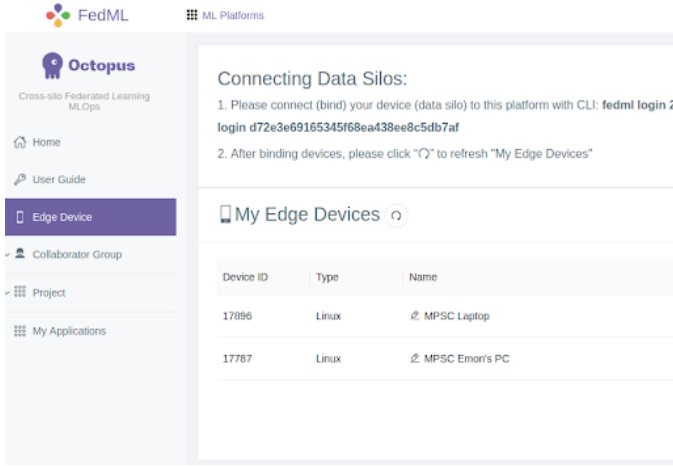


Fig. 4. FedML Platform MLOps Configuration

into. We executed the subsequent python files within the img2vec folder.

We then extracted the features using the off the shelf feature extractor img2vec. We wrote python scripts using the extractor to extract the features of random images and store them as numpy arrays. We created a script using the feature extractor to select 50 random images from each of the individual classes within the two datasets and calculate the image feature vector and store all 50 of them in an array. We then took the median vectors of the 50 and compared them to the median vector from each class with the similarity function of the img2vec library. We stored the output in a csv so we could compare the general similarity of the classes in combination with the t-sne plot clusters.

We then created a t-distributed stochastic neighbor embedding plot (t-sne plot) to visualize the clusters of ships based on their features. We then used the stored image feature vectors from the feature extractor code, to create a t-sne plot so we could visualize the clusters of similar boats. We did this for both the Seaships and ABOships datasets.

In order to determine the perplexity values, we created perplexity vs divergence graphs. The perplexity value affects the data complexly. If it is too low, the tendency is to cluster too many points together and not increase the distance between different clusters. If it is too high, the tendency is to increase the distance between the different clusters and not cluster similar points together. For this reason it is necessary to create perplexity vs divergence graphs. The divergence value quantifies the difference between two probability distributions. For this use the difference would be between two different clusters. We determined the minimum divergence before stabilization and take its perplexity value. We used the graph and tested multiple values for perplexity to find the ones that showed the clusters in a more defined way.

V. EXPERIMENTAL RESULTS/EVALUATION

We were able to crop the images into their folders based on the pixel value annotations (image showing the cropped

images and respective folders)

We also created the csv files that show the similarity comparison between each of the classes based on the image feature vectors for both Seaships and ABOships. The similarity csv for SeaShips is displayed in the table below.

Folder 1	Folder 2	Similarity
general_cargo_ship	ore_carrier	0.93283546
general_cargo_ship	passenger_ship	0.8316747
general_cargo_ship	bulk_cargo_carrier	0.94349533
general_cargo_ship	container_ship	0.8864378
general_cargo_ship	fishing_boat	0.872766
ore_carrier	general_cargo_ship	0.93283546
ore_carrier	passenger_ship	0.841764
ore_carrier	bulk_cargo_carrier	0.9535892
ore_carrier	container_ship	0.85502636
ore_carrier	fishing_boat	0.8576178
passenger_ship	general_cargo_ship	0.8316747
passenger_ship	ore_carrier	0.841764
passenger_ship	bulk_cargo_carrier	0.88599604
passenger_ship	container_ship	0.82786536
passenger_ship	fishing_boat	0.9359858
bulk_cargo_carrier	general_cargo_ship	0.94349533
bulk_cargo_carrier	ore_carrier	0.9535892
bulk_cargo_carrier	passenger_ship	0.88599604
bulk_cargo_carrier	container_ship	0.8940104
bulk_cargo_carrier	fishing_boat	0.91079116
container_ship	general_cargo_ship	0.8864378
container_ship	ore_carrier	0.85502636
container_ship	passenger_ship	0.82786536
container_ship	bulk_cargo_carrier	0.8940104
container_ship	fishing_boat	0.8628088
fishing_boat	general_cargo_ship	0.872766
fishing_boat	ore_carrier	0.8576178
fishing_boat	passenger_ship	0.9359858
fishing_boat	bulk_cargo_carrier	0.91079116
fishing_boat	container_ship	0.8628088

Part of the similarity csv for ABOships is displayed in the table below.

In order to determine the necessary perplexity values for the t-sne plot, we created perplexity versus divergence graphs for Seaships and ABOships

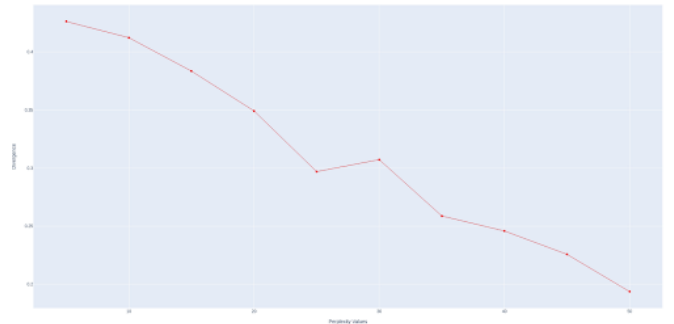


Fig. 5. ABOships Perplexity vs Divergence Graph

Finally, to visualize the clusters at the determined perplexity level we created a t-sne plot. We used perplexity value 17 for ABOships and 300 images from each class to cluster. For SeaShips we used a perplexity value of 25 and 70 images from each class to cluster.

Folder 1	Folder 2	Similarity
Boat	Miscellaneous	0.97630334
Boat	Passengership	0.8921411
Boat	Motorboat	0.97385085
Boat	Ferry	0.8302501
Boat	Militaryship	0.7427224
Boat	Miscboat	0.9201295
Boat	Cruiseship	0.77406377
Boat	Sailboat	0.6982449
Boat	Seamark	0.9686703
Boat	Cargoship	0.764315
Miscellaneous	Boat	0.97630334
Miscellaneous	Passengership	0.8277408
Miscellaneous	Motorboat	0.9299425
Miscellaneous	Ferry	0.774582
Miscellaneous	Militaryship	0.6884879
Miscellaneous	Miscboat	0.8579887
Miscellaneous	Cruiseship	0.72791207
Miscellaneous	Sailboat	0.65669966
Miscellaneous	Seamark	0.9677636
Miscellaneous	Cargoship	0.7039553
Passengership	Boat	0.8921411
Passengership	Miscellaneous	0.8277408
Passengership	Motorboat	0.96007264
Passengership	Ferry	0.93184316
Passengership	Militaryship	0.91633403
Passengership	Miscboat	0.9872719
Passengership	Cruiseship	0.91695833
Passengership	Sailboat	0.8418457
Passengership	Seamark	0.8748616
Passengership	Cargoship	0.9363429
Motorboat	Boat	0.97385085
Motorboat	Miscellaneous	0.9299425
Motorboat	Passengership	0.96007264
Motorboat	Ferry	0.8868901
Motorboat	Militaryship	0.8321693
...	...	...

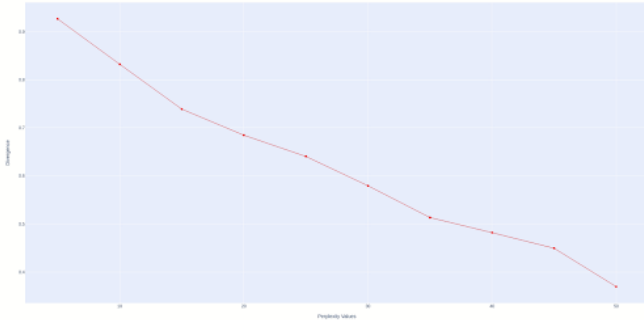


Fig. 6. SeaShips Perplexity vs Divergence Graph

VI. CONCLUSION

Throughout the paper we propose a framework and process for homogenizing heterogeneous datasets to train a federated model and determine necessary granularity for accurate model performance. We began the creation of a model for achieving statistical heterogeneity for a dataset through macro bucket image clustering. This method can be applied to dataset homogenization where there are annotation errors across many types of image datasets. It can also be used for clustering client data across the federated learning framework.

We can apply the methodology of feature extraction and

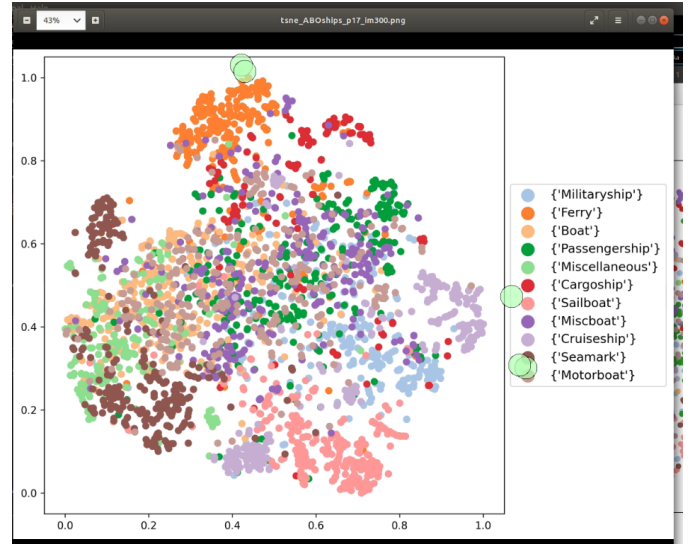


Fig. 7. ABOShips T-SNE Plot

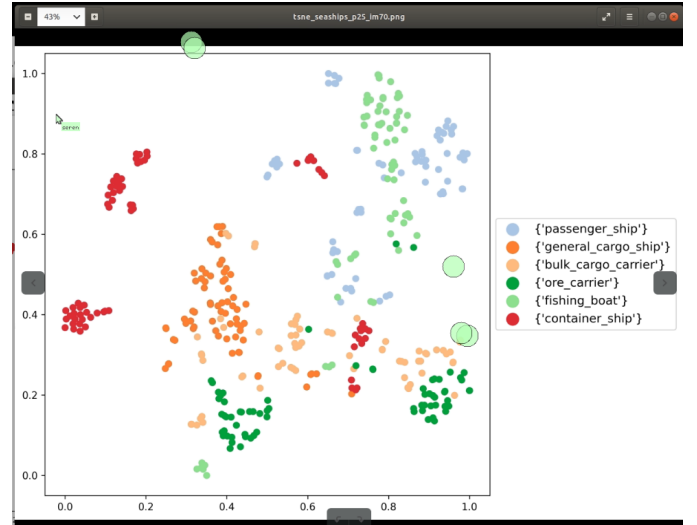


Fig. 8. SeaShips T-SNE Plot

clustering to a federated learning framework as well. In a federated learning setting the data of all of the clients would need to be made heterogeneous. Our proposed method for doing so to seaships datasets involving feature extraction and clustering could be applied to a federated learning environment.

Future research could involve using a clustering algorithm to compare and merge the classes of ships based on their extracted features and merging multiple datasets for training a federated learning model. Another approach can involve applying this process of feature extraction to a federated learning model and server clustering of its clients.

Throughout this process I learned the basics of machine learning and federated learning as well as python. I also learned to write and execute python code in an Ubuntu Linux terminal. I also learned to use repositories off of github and implement a feature extractor. I learned to read and write

to csv files and modify and sort files. I also learned about clustering and high dimensional modeling T-SNE plots and practiced creating them with code.

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VIII. ACKNOWLEDGEMENT

Thank you to UMBCs Department of Information Systems as well as Mr. Emon Dey, Dr. Ravi, and Dr. Roy for their support and guidance. This work has been partially supported by ONR Grant #N00014-23-1-2119, U.S. Army Grant #W911NF2120076, NSF CAREER Award #1750936, NSF REU Site Grant #2050999 and NSF CNS EAGER Grant #2233879.